

UCCD3094 Principles of Deep Learning

Assignment 2: Training / Finetuning Deep Learning Model

Trimester June 2026

Overview

In this project assignment, you are expected to finetune or train a deep learning model to solve a particular problem of your interest. Potential projects may fall into two tracks:

- **Application-based** - Train a deep learning model for a particular task of interest. Pick a real-world problem and apply deep learning techniques to solve it. Imagine that your team is competing against other teams to develop the *best* model for the application. To do that, you may want to:

- Perform *pre-processing* to prepare the data and make it suitable for your model, if possible.
- Apply *domain knowledge* to improve your system, if possible.
- Select the *suitable* network(s) for the task. You may use deep learning architectures or models beyond those covered in lectures. You are expected to demonstrate understanding of the architectures that you use *during the Q&A session*.
- *Finetune* your model (e.g., network architecture, hyperparameter settings, training methods etc.). **Your system must involve some form of training** (e.g., *training from scratch*,
- Do proper and *rigorous tests or experiments* (e.g., comparison with state-of-the-art models, qualitative evaluation, ablation study, etc.) to ensure your model is reliable and competitive.

Useful references:

- [Kaggle datasets](#): a collection of datasets published and shared by Kaggle users.
 - [Hugging Face Datasets](#): a collection of pre-processed and ready-to-use datasets for various NLP, computer vision and audio tasks.
 - [Past Stanford's CS231n Projects](#): Contains a list of projects from Stanford's CS21n course: Deep Convolutional Neural Networks for Visual Recognition.
- **Research-based.** Under this track, you may do one of following:
 - Study and *implement* some concepts, tasks or network architectures *not* covered in the lecture.
 - Study and *implement some recent and reputable research papers*¹.
 - Conduct *experiments* to study or understand the behavior of state-of-the-art systems.
 - *Improve* existing network or method.
 - Propose *novel* method for deep learning.

¹ By recent, it means the research paper is published within the recent 5 years. By reputable, it means that the paper is published in an [SCI-indexed](#) journal, had gone through a rigorous review process, and not a [predatory journal](#), or published in one of the [top-tier conference](#), e.g., CVPR, ICCV, NIPS, ICML, ICLR, ECCV, and AAAI.

For this category:

- The research problem or implemented system **should not be trivial**. For example, it is trivial to evaluate well established networks or methods (e.g., compare pretrained vs non-pretrained model, or the impact of L2 regularization), or re-implement classical neural network architectures such as AlexNet, VGGNet, ResNet, or GoogLeNet.
- We do not expect a full-fledged research-based project for this course. Novelty will be awarded but not a must for good grades.

General Rules

- Implement your system in **PyTorch**. Other frameworks are NOT allowed. However, **deep learning platform libraries or tool kit built on top of PyTorch (e.g., HuggingFace) are allowed**. You may implement your code in a notebook (.ipynb) or python file (.py), or both.
- You may use online pre-trained models in your assignments. In this case, your contribution comes from other parts, e.g., preprocessing, experiments or the deep learning system as a whole.
- You may reference papers, books, online resources, and public code for ideas, provided you cite all sources in your code and writeup. Sharing or copying another group's code is strictly prohibited.
- The work in this project should not be re-submitted to other courses for grading.
- Violation of rules may be subjected to a penalty depending on severity.

Use of Generative AI

- Generative AI may be used as a coding aid, but not to replace you as the developer. All AI use must be properly declared in the report.
- For the written report, AI can be used for editing, formatting, and proofreading, but not to replace you as the writer. The written component is intended to frame your thinking on the project; using AI to draft or write report content violates both the spirit of the assignment and the objectives of this course.
- **AI Use Declaration:** Include a brief statement at the end of your report listing any AI tools used, how (e.g., code generation, editing, formatting), and where (which function, class, module, or file) they were used for coding effort and report writing. **Provide links to relevant prompts or transcripts if available**. If no AI was used, state that explicitly.

Final report

- The **maximum number of pages is 10 pages** including references and AI use declaration. Pages beyond the maximum page may not be marked.
- You must write your report using the **provided template** (report_template.docx) which is attached together with this instruction sheet.

- The following is a suggested organization for your report. It is just a guideline. **You can have a different organization**, as long as the content is properly structured, clear and covers all important content.

Section	Description
Title and authors	List out all members of the team and specify the contribution of each author. If all members contribute equally, you may state so. State if the project is an application-based or research-based project.
Introduction	The purpose of this section is to provide an introduction and background for the project. The typical content in this section includes problem statement (what you want to solve), motivation of the project (why it is important) background of project (existing systems), challenges of the project (why it is difficult), and the project scope and objectives of the project. The introduction must be specific to your project. For example, if you are working on "car plate detection", you should <i>not</i> explain what "computer vision" or "deep learning" is. Rather, you should focus your writing on the task of detecting car plates and the related deep learning techniques (object detection).
Related Work (Optional, if any)	This section is optional. You can add this section if you wish to provide a more comprehensive review of existing works in the problem domain. Alternatively, you may discuss the related work when providing the background / context in the introduction section. For <i>research-based</i> projects, highlight the research gap in these works and tie them to your current project. There are many ways to write this part. You may review them from a chronological sense (historical perspective), or from the process perspective (break the system into different stages and discuss development in each stage), or by groups or categories (group related works based on techniques or methodologies), or from features (what the system are capable of doing) or a combination of these different approaches. For <i>application-based</i> project, you may review existing products or techniques and contrast them to your system.
System design or Implementation	This chapter discusses in detail how the project is developed. For both types of projects, you should give a <i>top-down system</i>

	<i>design</i> diagram to show your overall system (e.g., system flowchart, design block diagram). Then, explain in detail the functionality of each block in diagrams. Give sufficient details for others to re-implement your system. Avoid putting trivial implementation details (e.g., data structure or variable declaration, generic tools etc.). If your system implements some standard pre-trained model, you should introduce the model and provide the citation. However, you should not explain in detail well-established techniques, e.g., how convolution works, or the structure of an LSTM cell, etc. You may explain specialized or domain-specific techniques if it makes your writing clearer. The focus in this section is to write about “your project”. For <i>research-based</i> work, the main body of this section should describe your project at <i>concept-level</i> (e.g., the proposed / implemented network architecture, loss function or algorithms). Or if you are re-implementing an existing work, you need to explain the main idea behind the work (remember to cite!). Non-critical implementation details that is not related to your main research work (e.g., specific network architecture selected, training details, types of libraries used, etc.) can be written in a separate sub-section called "Implementation details". For <i>application-based</i> projects, the purpose is to describe how you build the system. You may organize your writing based on the machine learning pipeline (data collection, data preprocessing, network architecture used, training of the network). Any code that was used as a base for projects must be referenced and cited in the body of the paper.
Experiments and Evaluation	This purpose of section is to discuss the experiments that you have performed to demonstrate that your approach solves the problem. All results should be discussed in the main text with sufficient correct and objective <i>analysis</i>. Do not assume that the reader will be able to interpret the results by themselves. - Verify the effectiveness of your system, use suitable performance metric to validate your system. - Perform hyperparameter tuning to achieve best performance for your model. - Others:

	○ You may use visualization techniques to gain insight into how your model works, discuss common failure modes of your model, etc. ○ You may <i>compare</i> the performance of your model between different network models. The comparison ○ You may perform ablation studies to evaluate the contribution of individual components on your system. ○ Any other evaluations that allow better understanding of your system.
Conclusion	Summarize your key results - what have you learned? Suggest ideas for future extensions or new applications of your ideas.
AI Use Declaration	Include a brief statement at the end of your report listing any AI tools used, how (e.g., code generation, editing, formatting), and where (which function, class, module, or file) they were used for coding effort and report writing. Provide links to relevant prompts or transcripts if available. If no AI was used, state that explicitly.
Reference	Cite all the works that you refer to. All references that you include in this section must be cited in the main text. For this project, we shall use the same reference style as FYP projects. The IEEE citation format can be found in this guide .

Team Formation

- All teams must have **4 members** except for Team 19 which has **3 members**. Team members can be from any practical group. Register the team members for your team in this [Google spreadsheet](#).

Project Submission

- Submission date: **Week 12, Sunday (6 September 2026)**
- Only **one person** on your team should submit to WBLE. Please put the names of your teammates into your report.
- The submission is a **single zip file** with the name *Assignment2_group<ID>_<LeaderName>.zip*.
 - **All file size is limited to 5MB.**
 - **Do not** put your dataset or model into the file. You should put them into your own online drive or repository. Provide the link in the read me file and erase them only after your result is announced.
- The contents in the report submission include:
 - **Final report** (.pdf file)
 - **Implementation:**
 - **Source codes** (required, either .ipynb or .py files)

- **Readme file** (optional) (e.g., description of submitted files, how to run the system, link to dataset, model, video or demo, if any and other information.

Presentation & Demo

- All teams are required to submit the link to your **presentation video** and **power-point file** to WBLE by **Week 13 Wednesday (9 Sept 2026)**.
- Presentation duration is limited to a maximum duration of **25 minutes**. All team members must present in the video. The presentation should have the following contents:
 1. Background of the project (problem statement)
 2. Project objective and scope
 3. Methodology
 4. Experimental results
 5. **System demo** (show what your trained model can do, and the developed interface, if any)

Marking Scheme & Rubric

System Development		40%
	Problem statement & formulation	25%
	Implementation & Prototyping	15%
Experiment & Analysis		35%
	Experiment & Analysis	35%
Report & Presentation		20%
	Report writing	10%
	Presentation	10%
Others		5%
	General effort and Work ethics.	5%

Criteria	Very Good	Good	Average	Poor	Not Acceptable
Problem Formulation & Modeling	Problem statement is excellently explained	Problem statement is well explained.	Problem statement is sufficiently clear.	Problem statement is not clear.	Problem statement is not defined
	Model is appropriate.	Model is appropriate.	Model is appropriate.	Model is appropriate.	Model is not appropriate.
	System design is complex.	System design is somewhat complex.	Simple customization of the model	Direct application of standard pre-trained model.	Wrong system design.
Implementation & prototyping	System is working fully functional with excellent capabilities.	System is fully functional with some interesting capabilities.	System is fully functional.	System is partially functional.	System is not working.

	Implementation is comprehensive and complex	Implementation is somewhat complex.	Implementation is at adequate level.	Implementation is sloppy and bareboned.	No prototype, or prototype is similar to online repository.
Experiment & Analysis	Excellent documentation on experimental setup and dataset description.	Good documentation on experimental setup and dataset description.	Simple description of dataset and experimental setup provided.	Insufficient description of dataset and experimental setup.	No description of dataset and experimental setup
	Elaborate and effective hyper-parameter finetuning performed.	Effective hyper-parameter finetuning performed.	Basic hyper-parameter finetuning performed.	No evidence of hyper-parameter finetuning performed.	No evidence of hyper-parameter finetuning performed.
	In-depth and insightful analysis made on the experimental result.	Useful analysis made on the experimental result.	Simple analysis made on the experimental result.	Mainly reporting the numbers in the experimental results.	No analysis provided.
	Performance comparison with SOTA models. Insightful ablation study conducted.	Performance comparison with other strong models. Some ablation study conducted.	Performance comparison with some models. No ablation study conducted.	Limited performance comparison with other models. No ablation study.	No performance comparison with other models.No ablation study conducted.
Report writing	Problem statement objective and scope are clearly and concisely stated.	Problem statement, project scope and objectives are clear though not as concise or comprehensive.	Problem statement, project scope and objectives are stated and understandable	Problem statement and objectives are unclear or incomplete.	No clear problem statement, objective, or scope is provided.
	System architecture and implementation clearly explained with sufficient detail and clarity.	System architecture and implementation explained and understandable, but not fully detailed.	System architecture or implementation is minimally described but still present.	Explanation is vague or confusing, with important details missing.	Lacking explanation of system architecture or implementation.
	Experimental setup is clear. Results completely and clearly analyzed and interpreted.	Experimental setup provided. Results are analyzed but interpretation not fully comprehensive.	Experimental setup present but basic. Results minimally analyzed but present.	Results are superficially presented with little or no meaningful analysis.	Lacking experimental setup, results, or analysis provided.
Presentation	Exceptional presentation. Easy to follow, well-structured and interesting.	Good presentation. Easy to follow and well structured.	Acceptable presentation. Understandable.	Poor presentation. Difficult to follow and unorganized.	No presentation
	Covers all aspect of the project well.	Covers all aspects of the project satisfactorily.	Covers all aspect of the project adequately.	Covers critical aspect of the project adequately.	Does not cover critical aspect f the project.
	Excellent slide design, amount of text is just right, excellent use of visual aids.	Good slide design, generally not verbose, effective use of visual aids.	Acceptable slide design, slightly verbose, some use of visual aids.	Poor slide design – verbose and lacking visual aids.	No slide used to present

General Effort & Work Ethics	Hardworking and putting a lot of thought into the project. Satisfies most of the rules & regulations.	Hardworking. Satisfies most rules & regulations.	Satisfies most rules & regulations.	Break major rules & regulations.	Break major rules & regulations
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